CFD analysis of the aerodynamic performance of NACA 2412 unsymmetrical airfoil

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CERTIFICATE

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Abstract

Computational Fluid Dynamics (CFD) analysis of the aerodynamic performance of different symmetrical and unsymmetrical airfoils under subsonic flow conditions. The primary objective is to compare key aerodynamic parameters such as lift coefficient (CL), drag coefficient (CD), pressure distribution, and lift-to-drag ratio (L/D) at various angles of attack. Numerical simulations are carried out by solving the Navier–Stokes equations using an appropriate turbulence model, with refined meshing near the airfoil surface to accurately capture boundary-layer effects. The results demonstrate that symmetrical airfoils produce zero lift at zero angle of attack and exhibit stable performance over a wide operating range, whereas unsymmetrical (cambered) airfoils generate higher lift and improved aerodynamic efficiency at lower angles of attack. Flow visualization through velocity contours, pressure plots, and streamlines highlights the influence of airfoil geometry on flow separation and stall behavior. The comparative analysis provides valuable insights into airfoil selection and optimization for aerospace, wind energy, and related engineering applications.

Keywords: Angle of Attack;Drag Coefficient (CD); Lift Coefficient (CL); Pressure Distribution

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Abbreviations

Abbreviation	Description
VCE	Vardhaman College of Engineering
CFD	Computational Fluid Dynamics

CHAPTER 1

Introduction

1.1 Importance of airfoil analysis in aerospace/wind energy

Airfoil analysis plays a crucial role in both aerospace engineering and wind energy systems because it directly determines how efficiently a structure interacts with airflow. In aerospace applications, the performance of wings, propellers, and turbine blades depends on how well an airfoil generates lift while minimizing drag. Through analysis—often using Computational Fluid Dynamics (CFD) or wind tunnel testing—engineers can predict aerodynamic forces, optimize shapes, and ensure stable flight conditions. This is especially important for aircraft safety, fuel efficiency, and performance under varying speeds and atmospheric conditions. A well-designed airfoil improves lift-to-drag ratio, enabling aircraft to fly longer distances with less fuel consumption.

In wind energy, airfoil analysis is equally important because wind turbine blades essentially function as rotating airfoils. Their ability to extract energy from the wind depends on aerodynamic efficiency. By analyzing airfoil behavior at different wind speeds and angles of attack, engineers can design blades that maximize power output while reducing structural loads and fatigue. Efficient airfoil designs lead to higher energy conversion efficiency, making renewable energy systems more cost-effective and sustainable.

Additionally, airfoil analysis helps in understanding critical phenomena such as flow separation, stall, and turbulence. Preventing early stall and reducing aerodynamic losses ensures smoother operation in both aircraft and wind turbines. It also supports noise reduction and durability improvements, which are essential for environmental compliance and long-term reliability.

1.2 Applications

Airfoils are widely used in many engineering applications because of their ability to efficiently generate lift and control airflow. Their design and analysis are essential in systems where aerodynamic performance directly affects efficiency, stability, and functionality.

Aircraft Wings: In aircraft, airfoils form the primary shape of wings and are responsible for generating lift, which allows the aircraft to fly. The curvature and thickness of the airfoil determine how air flows over the wing, influencing lift and drag forces. Modern airplanes use carefully optimized airfoils to achieve high lift-to-drag ratios, ensuring fuel efficiency, smooth flight, and safety. Different types of airfoils are used depending on the aircraft's purpose—commercial planes focus on efficiency, while fighter jets use specialized airfoils for high-speed maneuverability.

Wind Turbines: Wind turbine blades are designed using airfoil principles to convert wind energy into mechanical energy. As wind passes over the blades, lift is generated, causing the rotor to spin. Advanced airfoil designs help maximize energy extraction even at low wind speeds while minimizing losses due to drag. Airfoil analysis also helps reduce fatigue loads on blades, increasing their lifespan and improving the reliability of wind energy systems.

Drones (UAVs): In drones or unmanned aerial vehicles, airfoils are used in wings and propellers to ensure stable and efficient flight. Since drones often operate at lower speeds and varying conditions, their airfoil design is optimized for better control, lift generation, and energy efficiency. Efficient airfoils help drones achieve longer flight times, better payload capacity, and improved maneuverability, which is especially important in applications like surveillance, delivery, and mapping..

1.3 Brief introduction to Computational Fluid Dynamics

Computational Fluid Dynamics (CFD) is a branch of Fluid Mechanics that uses numerical methods and algorithms to analyze and simulate the behavior of fluids (liquids and gases) and their interaction with surfaces. Instead of relying only on physical experiments, CFD allows engineers to study complex flow patterns using computers, making analysis faster, more cost-effective, and highly detailed.

At the core of CFD are mathematical equations known as the Navier–Stokes equations, which describe how fluid velocity, pressure, temperature, and density change over time and space. These equations are solved numerically by dividing the flow domain into small control volumes (a process called meshing) and applying appropriate boundary conditions.

CFD is widely used in many engineering fields. In aerospace, it helps analyze airflow over wings and predict lift and drag forces. In automotive engineering, it is used to improve vehicle aerodynamics and cooling systems. In energy applications, such as wind turbines, CFD helps optimize blade design for maximum efficiency. It is also used in environmental studies, biomedical engineering, and industrial processes.

1.4 Problem statement and motivation

Problem Statement: In modern engineering applications such as aircraft, wind turbines, and drones, achieving high aerodynamic efficiency is a major challenge. Airfoils must generate sufficient lift while minimizing drag under a wide range of operating conditions. However, variations in angle of attack, flow velocity, and environmental conditions often lead to issues such as flow separation, increased drag, and early stall. These problems reduce performance, increase energy consumption, and may even affect safety and stability. Traditional experimental methods like wind tunnel testing are costly and time-consuming, making it difficult to analyze multiple airfoil designs efficiently.

Therefore, there is a need for an accurate, cost-effective, and reliable approach to study and optimize the aerodynamic performance of both symmetrical and unsymmetrical airfoils under different flow conditions.

Motivation: The motivation behind this study is to improve the understanding and performance of airfoils using advanced simulation techniques like Computational Fluid Dynamics. By applying CFD, engineers can visualize airflow behavior, predict lift and drag characteristics, and identify critical phenomena such as stall and turbulence without extensive physical testing. This helps in designing more efficient aircraft wings, increasing power output in wind turbines, and enhancing the flight performance of drones. Additionally, improving aerodynamic efficiency contributes to reduced fuel consumption, lower operational costs, and minimized environmental impact. Hence, this study is driven by the need to develop optimized airfoil designs that enhance performance, efficiency, and sustainability in real-world applications.

1.5 Organization of the Report

Chapter 1: Introduction Provides an overview of the project, including background, motivation, objectives, and significance of the study.

Chapter 2: Literature Review Discusses previous research and studies related to airfoil analysis and Computational Fluid Dynamics (CFD).

Chapter 3: Methodology Describes the CFD modeling process, including geometry creation, mesh generation, governing equations, boundary conditions, and solver setup.

Chapter 4: Results and Discussion Presents the simulation results such as lift and drag coefficients, pressure distribution, and flow visualization. A comparative analysis of symmetrical and unsymmetrical airfoils is also included.

Chapter 5: Conclusion and Future Work Summarizes the key findings of the study and suggests possible improvements and future research directions.

Bibliography

[1] I. H. Abbott and A. E. Von Doenhoff, *Theory of Wing Sections: Including a Summary of Airfoil Data*. New York, NY, USA: Dover Publications, 1959.

[2] J. D. Anderson, *Fundamentals of Aerodynamics*, 6th ed. New York, NY, USA: McGraw-Hill, 2017.

[3] R. Ives, A. S. Keir, E. Bassey, and F. A. Hamad, "Investigation of the flow around an aircraft wing of section NACA 2412 utilising ANSYS Fluent," *INCAS Bulletin*, vol. 10, no. 1, pp. 95–104, 2018.

[4] M. S. G. Samuel, P. Rajendran, and S. A. Khan, "CFD simulation of NACA 2412 airfoil with new cavity shapes," *Advances in Aircraft and Spacecraft Science*, vol. 9, no. 2, pp. 131–148, 2022.

[5] Z. Gong, "CFD analysis of NACA2412 and NACA0012 at low Reynolds numbers based on ANSYS Fluent," *Applied and Computational Engineering*, vol. 127, no. 1, pp. 59–67, 2025.

CHAPTER 2

Literature Survey

2.1 Introduction

Airfoils are fundamental components in aerodynamic systems such as aircraft wings, unmanned aerial vehicles (UAVs), wind turbines, and turbomachinery, where their shape significantly influences lift, drag, and overall performance. Airfoils are generally classified into symmetrical and unsymmetrical (cambered) types, each exhibiting different aerodynamic characteristics depending on operating conditions.

With the advancement of computational techniques, Computational Fluid Dynamics (CFD) has become an effective tool for analyzing airflow behavior around airfoils. By solving the governing fluid flow equations, CFD enables the prediction of important aerodynamic parameters such as lift coefficient (C), drag coefficient (C), pressure distribution, and flow patterns without the need for costly experimental methods.

2.2 Review of Prior Research

Several researchers have studied the aerodynamic performance of airfoils using both experimental methods and Computational Fluid Dynamics (CFD) techniques. Early studies using wind tunnel experiments provided valuable data on lift, drag, and pressure distribution, but these methods were often expensive and time-consuming.

With the development of CFD, many studies have focused on analyzing airfoil behavior under different flow conditions. Researchers have shown that CFD can accurately predict aerodynamic parameters such as lift coefficient (C), drag coefficient (C), and flow separation, making it a reliable alternative to experimental methods. Various turbulence models, including the k– and k– models, have been widely used to simulate turbulent flow and improve prediction accuracy.

Comparative studies between symmetrical and unsymmetrical airfoils have indicated that symmetrical airfoils produce zero lift at zero angle of attack and are suitable for applications requiring stability. In contrast, cambered airfoils generate higher lift at lower angles of attack, making them more efficient for lifting applications such as aircraft wings and wind turbine blades.

2.3 Identified Research Gaps

Many studies focus on either symmetrical or unsymmetrical airfoils individually, with limited direct comparative analysis under identical conditions.

The effect of angle of attack on both airfoil types is not consistently analyzed over a wide operating range, especially near stall conditions.

Some research lacks detailed investigation of flow separation and stall behavior using advanced flow visualization techniques such as velocity contours and streamlines.

Limited emphasis is given to the combined analysis of lift, drag, and lift-to-drag ratio (L/D) for performance optimization.

2.4 Summary

This chapter reviewed the fundamental concepts and previous research related to the aerodynamic performance of symmetrical and unsymmetrical airfoils using Computational Fluid Dynamics (CFD). It highlighted the importance of airfoil design in applications such as aerospace, UAVs, and wind energy systems. The literature survey showed that CFD is an effective tool for analyzing parameters like lift, drag, pressure distribution, and flow behavior.

The review also identified key differences between symmetrical and cambered airfoils, along with the influence of angle of attack on performance and stall characteristics. Furthermore, several research gaps were identified, including the need for detailed comparative analysis under consistent conditions and improved flow visualization studies.

Bibliography

[1] R. Ives, A. S. Keir, E. Bassey, and F. A. Hamad, "Investigation of the flow around an aircraft wing of section NACA 2412 utilising ANSYS Fluent," *INCAS Bulletin*, vol. 10, no. 1, pp. 95–104, 2018.

[2] M. S. G. Samuel and P. Rajendran, "CFD validation of NACA 2412 airfoil," Technical Report, 2019.

[3] M. S. G. Samuel, P. Rajendran, and S. A. Khan, "CFD simulation of NACA 2412 airfoil with new cavity shapes," *Advances in Aircraft and Spacecraft Science*, vol. 9, no. 2, pp. 131–148, 2022.

[4] Z. Gong, "CFD analysis of NACA2412 and NACA0012 at low Reynolds numbers based on ANSYS Fluent," *Applied and Computational Engineering*, vol. 127, no. 1, pp. 59–67, 2025.

[5] N. D. Baruah, D. Deka, and K. K. Das, "Aerodynamic analysis of NACA 2412 airfoil using ANSYS," *International Research Journal on Advanced Engineering and Management*, vol. 3, no. 2, pp. 173–177, 2025.

[6] M. R. Robby, A. Karudin, Refdinal, and Y. Fernanda, "Aerodynamic analysis of NACA 2412 airfoil on UAV aircraft wing using ANSYS Fluent software," *International Journal of Innovation in Mechanical Construction and Energy*, vol. 1, no. 3, 2024.

[7] Y. Guo, "Aerodynamic performance and flow characteristics of the 2D NACA 2412 airfoil using CFD," *Applied and Computational Engineering*, 2026.

CHAPTER 3

Geometry Modelling and Analysis

3.1 Airfoil selection (NACA 2412)

The selection of an appropriate airfoil is a crucial step in aerodynamic analysis, as it directly affects lift, drag, and overall performance. In this study, the NACA 2412 airfoil is chosen due to its well-balanced aerodynamic characteristics and wide usage in practical applications.

The NACA 2412 is a four-digit airfoil defined by the National Advisory Committee for Aeronautics classification system. Each digit has a specific meaning: the first digit (2) represents a maximum camber of 2

One of the main reasons for selecting NACA 2412 is its good lift-to-drag ratio and smooth stall characteristics. It provides stable aerodynamic performance over a wide range of operating conditions, which is particularly useful in applications such as light aircraft, wind turbines, and unmanned aerial vehicles. Compared to symmetrical airfoils, NACA 2412 offers higher lift generation due to its cambered shape, improving overall efficiency.

3.2 Coordinate generation (Airfoil tools or databases)

The generation of airfoil coordinates is an essential step in geometry modelling, as it provides the precise shape needed for aerodynamic analysis. For the NACA 2412 airfoil, coordinates define the upper and lower surface profiles along the chord length, ensuring an accurate representation of its camber and thickness distribution.

Airfoil coordinates can be obtained from reliable online databases and tools such as Airfoil Tools or the UIUC Airfoil Database. These platforms provide pre-generated coordinate data for a wide range of airfoils, including NACA series. The data is typically available in a simple text format consisting of x (chord-wise position) and y (vertical position) values, which can be directly imported into CAD or CFD software.

Alternatively, coordinates can also be generated mathematically using standard NACA equations. For a 4-digit airfoil like NACA 2412, the camber line and thickness distribution are calculated using empirical formulas, and the final coordinates are derived by combining these values. This approach is useful when customized airfoil shapes or higher precision is required.

Once the coordinate data is obtained, it is imported into modeling software such as ANSYS or SolidWorks. The points are then connected to form smooth curves representing the airfoil profile. Care must be taken to ensure proper point ordering (from leading edge to trailing edge) and sufficient resolution for accurate analysis.

3.3 Domain creation

Domain creation is a critical step in Computational Fluid Dynamics (CFD) analysis, where the fluid region surrounding the airfoil is defined for simulation. Instead of analyzing only the airfoil surface, a larger computational domain is created to represent the airflow environment around it. This ensures that the simulation captures realistic flow behavior without interference from artificial boundaries.

For the NACA 2412 airfoil, the domain is typically designed as a 2D rectangular or C-type region. The airfoil is placed inside this domain, usually with sufficient distance from all boundaries to avoid boundary effects. Standard practice is to keep the inlet boundary about 5–10 chord lengths upstream of the airfoil, the outlet boundary about 10–15 chord lengths downstream, and the top and bottom boundaries around 5–10 chord lengths away. This spacing ensures that the airflow develops naturally before interacting with the airfoil and exits without affecting the results.

The domain is divided into different boundary regions such as:

Inlet: where airflow enters with a specified velocity

Outlet: where pressure is defined and flow exits

Top and Bottom Walls: often treated as far-field or symmetry conditions

Airfoil Surface: defined as a no-slip wall condition

The domain shape can vary depending on the type of analysis. A rectangular domain is simple and widely used, while a C-type or O-type domain provides better accuracy near the airfoil by allowing smoother mesh distribution around curved surfaces.

3.4 Boundary conditions

CBoundary conditions are essential in Computational Fluid Dynamics as they define how the fluid behaves at the limits of the computational domain. Proper specification of these conditions ensures that the simulation accurately represents real physical flow around the airfoil.

For the NACA 2412 airfoil, the commonly applied boundary conditions are as follows:

Inlet (Velocity Inlet): At the inlet boundary, a uniform airflow velocity is specified along with flow direction. This represents the free-stream conditions approaching the airfoil. Parameters such as turbulence intensity and viscosity may also be defined depending on the model used.

Outlet (Pressure Outlet): At the outlet, static pressure is specified (usually atmospheric pressure). This allows the fluid to exit the domain naturally without reflecting back into the flow field.

Airfoil Surface (Wall Condition): The airfoil surface is treated as a no-slip wall, meaning the fluid velocity at the surface is zero. This condition is important for accurately capturing the boundary layer and shear effects.

Top and Bottom Boundaries (Far-Field or Symmetry): These boundaries are placed far from the airfoil and are often defined as symmetry or free-stream conditions. This ensures that they do not disturb the airflow around the airfoil.

Angle of Attack (AoA): The angle of attack is implemented either by changing the flow direction at the inlet or by rotating the airfoil geometry. This parameter is critical as it directly affects lift and drag characteristics.

3.5 Mesh Generation

Mesh generation is a vital step in Computational Fluid Dynamics, where the computational domain around the airfoil is divided into small discrete elements or cells. These cells are used to numerically solve the governing equations of fluid flow. The accuracy of the simulation largely depends on the quality and refinement of the mesh.

For the NACA 2412 airfoil, a structured or unstructured mesh can be generated depending on the complexity of the domain. Structured meshes (like quadrilateral grids) provide better accuracy and smoother flow resolution, while unstructured meshes (triangular elements) offer flexibility in handling complex geometries. In many cases, a hybrid mesh is used to balance accuracy and computational cost.

A key aspect of mesh generation is refinement near the airfoil surface. The region close to the wall, known as the boundary layer, experiences steep velocity gradients. To capture this accurately, finer mesh elements and inflation layers are applied along the airfoil surface. These layers help resolve shear stress and predict drag more precisely.

3.6 Setup

The simulation setup is a crucial stage in Computational Fluid Dynamics, where all physical models, numerical methods, and solution parameters are defined before solving the flow problem. A proper setup ensures accurate and stable results for the aerodynamic analysis of the NACA 2412 airfoil.

First, the solver type is selected. For airfoil analysis, a pressure-based solver is commonly used, assuming steady, incompressible flow for low-speed conditions. If the flow involves high speeds, a compressible solver may be required.

Next, the turbulence model is chosen to capture real flow behavior. Models like Spalart–Allmaras or k– are widely used for external aerodynamic simulations. The Spalart–Allmaras model is particularly suitable for airfoil analysis due to its efficiency and good prediction of boundary layer flows.

The material properties of the fluid (air) are then defined, including density and viscosity. After this, previously defined boundary conditions (velocity inlet, pressure outlet, wall conditions) are applied to the domain.

The solution methods are set by selecting discretization schemes (such as second-order upwind) for better accuracy. Pressure–velocity coupling methods like SIMPLE are also used to ensure convergence of the solution.

Following this, initial conditions are provided, typically assuming uniform flow at the inlet velocity. The solution is then iterated until convergence is achieved. Convergence is monitored using residuals (errors) and stability of key parameters like lift and drag coefficients.

Finally, solution controls such as relaxation factors and number of iterations are adjusted to improve stability and accuracy.

CHAPTER 4

Results and Discussion

4.1 Static pressure contours of airfoil

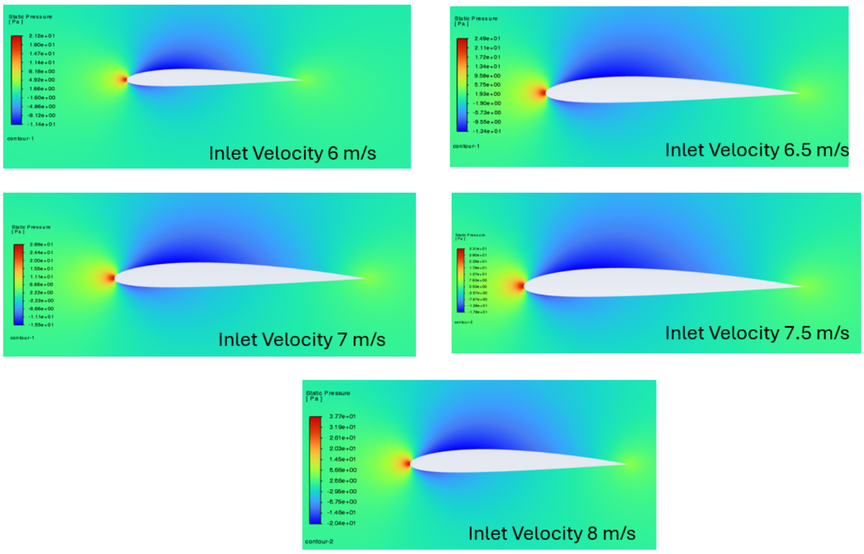

Figure 4.1: Static pressure contours of airfoil at different inlet velocities

The CFD analysis of the airfoil was conducted at inlet velocities of 6 m/s, 6.5 m/s, 7 m/s, 7.5 m/s, and 8 m/s to study the variation in static pressure distribution and its aerodynamic effects, as shown in Figure 4.1. The pressure contour results show a consistent aerodynamic pattern across all cases, with a high-pressure region forming at the leading edge due to the stagnation effect and a low-pressure region developing over the upper surface of the airfoil. This pressure difference between the upper and lower surfaces is responsible for the generation of lift. As the flow moves towards the trailing edge, the pressure gradually recovers, indicating smooth and attached flow behavior.

With an increase in inlet velocity, a significant change in pressure magnitude and distribution is observed, as summarized in Table 4.1. At lower velocities such as 6 m/s and 6.5 m/s, the pressure difference is relatively moderate, resulting in lower lift generation. As the velocity increases to 7 m/s and 7.5 m/s, the suction effect on the upper surface becomes more prominent, and the pressure gradient between the upper and lower surfaces increases. At the highest velocity of 8 m/s, the pressure contours shown in Figure 4.1 indicate the strongest suction region and maximum pressure difference, suggesting the highest lift potential among all cases.

In addition to lift, drag characteristics also vary with velocity, as presented in Table 4.1. As the inlet velocity increases, the overall drag force on the airfoil also increases. This is primarily due to the increase in dynamic pressure and shear effects. The pressure drag component increases slightly due to higher pressure differences, while the skin friction drag increases due to higher velocity gradients near the airfoil surface. However, since no significant flow separation is observed in any of the cases in Figure 4.1, the pressure drag remains relatively controlled. The smooth pressure recovery towards the trailing edge indicates that the airfoil maintains efficient aerodynamic performance with minimal wake formation.

Although drag increases with velocity, the rate of increase in lift is comparatively higher, leading to an improved lift-to-drag ratio at higher velocities, as evident from Table 4.1. Overall, the results demonstrate that increasing inlet velocity enhances both lift and drag forces, but the aerodynamic efficiency of the airfoil improves due to a greater increase in lift compared to drag. These findings are consistent with fundamental aerodynamic principles and validate the performance of the airfoil within the tested velocity range.

Table 4.1: Comparison of Aerodynamic Performance at Different Velocities

Velocity (m/s)	**Pressure Difference**	**Lift**	**Drag**	**Flow Behavior**
6	Low	Low	Low	Smooth
6.5	Moderate	Moderate	Slight	Smooth
7	High	High	Moderate	Attached
7.5	Very High	Very High	High	Stable
8	Maximum	Maximum	Highest	Efficient

4.2 Turbulent Kinetic Energy counters of airfoil

The CFD analysis of the airfoil was carried out at inlet velocities of 6 m/s, 6.5 m/s, 7 m/s, 7.5 m/s, and 8 m/s to study the variation of turbulent

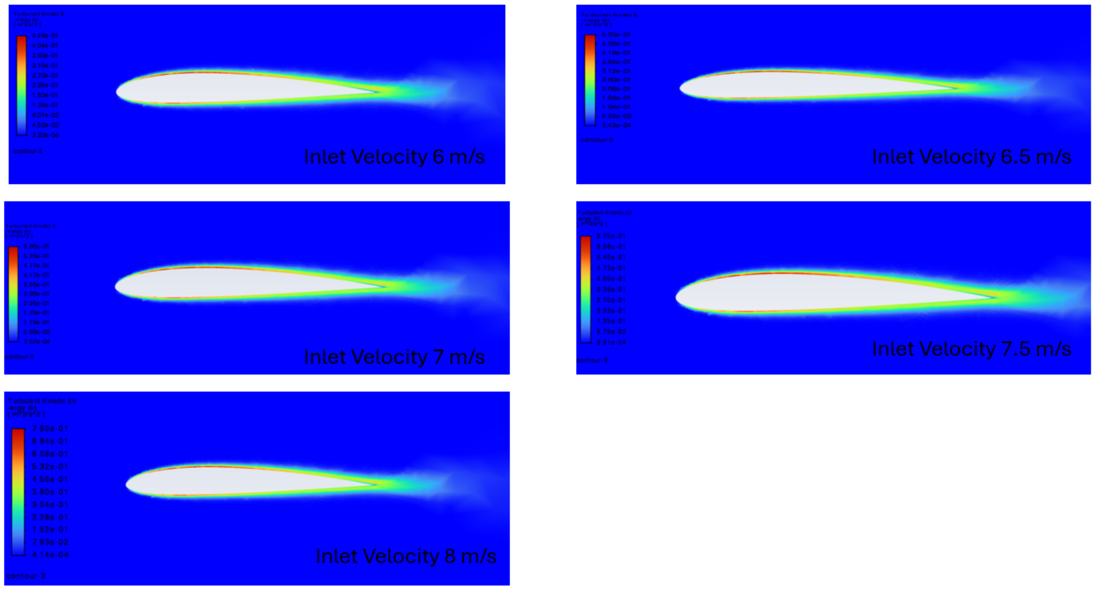

Figure 4.2: Static pressure contours of airfoil at different inlet velocities

kinetic energy (TKE) around the airfoil, as shown in Figure 4.2. The contour plots indicate that the turbulent kinetic energy is mainly concentrated near the airfoil surface and in the wake region behind the trailing edge. The leading edge shows relatively low turbulence levels, while turbulence gradually develops along the airfoil surface and increases significantly towards the trailing edge.

As the inlet velocity increases, the magnitude of turbulent kinetic energy also increases, which can be observed clearly in Figure 4.2 and summarized in Table 4.2. At lower velocities such as 6 m/s and 6.5 m/s, the turbulence levels are relatively low, and the wake region is less intense. As the velocity increases to 7 m/s and 7.5 m/s, the turbulent kinetic energy becomes more prominent, especially near the trailing edge where flow separation tendencies begin to increase slightly. At the highest velocity of 8 m/s, the contours in Figure 4.2 show the maximum turbulence intensity, along with a more extended and stronger wake region behind the airfoil.

The increase in turbulent kinetic energy with velocity is mainly due to higher velocity gradients and increased shear effects in the boundary layer. The trailing edge region experiences the highest turbulence due to flow mixing and wake formation. Despite the increase in turbulence, the flow remains largely attached over the airfoil surface in all cases shown in Figure 4.2, indicating stable aerodynamic behavior within the tested range.

Overall, the results indicate that higher inlet velocities lead to increased turbulence intensity and wake formation, which contribute to higher drag, as summarized in Table 4.2. However, since the turbulence is mainly confined to the wake region and does not cause major flow separation, the airfoil continues to maintain efficient aerodynamic performance over the tested velocity range.

Table 4.2: Comparison of Turbulent Kinetic Energy at Different Velocities

Velocity (m/s)	**Turbulence Intensity**	**Wake Strength**	**Flow Behavior**	**Drag Effect**
6	Low	Weak	Smooth	Low
6.5	Moderate	Slight	Smooth	Slight
7	High	Noticeable	Attached	Moderate
7.5	Very High	Strong	Stable	High
8	Maximum	Very Strong	Stable	Highest

4.3 Velocity Magnitude contours of airfoil

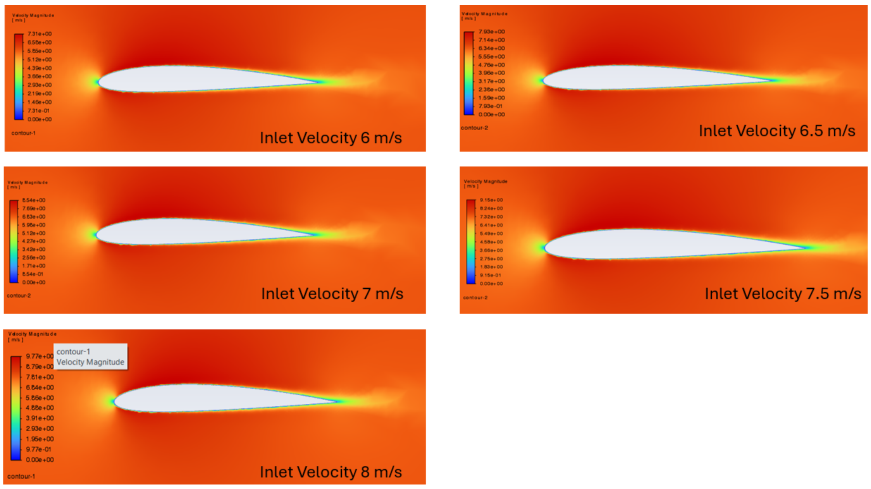

Figure 4.3: Streamline patterns of airfoil at different inlet velocities

The CFD analysis of the airfoil was performed at inlet velocities of 6 m/s, 6.5 m/s, 7 m/s, 7.5 m/s, and 8 m/s to study the variation of velocity magnitude around the airfoil, as shown in Figure 4.3. The velocity contour plots indicate that the flow accelerates over the upper surface of the airfoil and decelerates near the leading edge, forming a stagnation region. The maximum velocity is observed over the upper surface due to flow acceleration, while the wake region behind the trailing edge shows reduced velocity.

As the inlet velocity increases, the overall velocity magnitude across the flow field also increases proportionally, which is clearly observed in Figure 4.3 and summarized in Table 4.3. At lower velocities such as 6 m/s and 6.5 m/s, the acceleration over the airfoil surface is moderate, and the wake region is relatively small. As the velocity increases to 7 m/s and 7.5 m/s, the flow acceleration becomes more significant, leading to higher velocity gradients over the airfoil surface. At the highest velocity of 8 m/s, the contours in Figure 4.3 show the maximum velocity magnitude, along with a more pronounced wake region behind the trailing edge.

The increase in velocity over the upper surface corresponds to a decrease in pressure, which supports lift generation according to Bernoulli's principle. The wake region becomes slightly larger at higher velocities due to increased shear and mixing effects. However, the flow remains attached over the airfoil surface in all cases shown in Figure 4.3, indicating stable aerodynamic performance without significant flow separation.

Overall, the results indicate that increasing inlet velocity enhances flow acceleration and velocity gradients around the airfoil, thereby contributing to higher lift generation. Although the wake region increases slightly, the airfoil maintains efficient aerodynamic behavior within the tested velocity range, as summarized in Table 4.3.

Table 4.3: Comparison of Velocity Magnitude at Different Velocities

Velocity (m/s)	**Maximum Velocity**	**Flow Acceleration**	**Wake Region**	**Flow Behavior**
6	Moderate	Moderate	Small	Smooth
6.5	Slightly High	Increased	Slight	Smooth
7	High	Strong	Noticeable	Attached
7.5	Very High	Very Strong	Larger	Stable
8	Maximum	Maximum	Largest	Stable

4.4 Streamline Patterns

The CFD analysis of the airfoil was further extended to study the flow behavior using velocity streamlines at inlet velocities of 6 m/s, 6.5 m/s, 7 m/s, 7.5 m/s, and 8 m/s, as shown in Figure 4.4. The streamline plots provide a clear visualization of the flow pattern around the airfoil, illustrating how the fluid moves over and around the airfoil surface.

At all inlet velocities, the streamlines smoothly approach the leading edge and divide into upper and lower flow paths, indicating proper flow attachment. Over the upper surface of the airfoil, the streamlines are closely spaced, representing accelerated flow, while on the lower surface, the spacing is relatively wider, indicating lower velocity. This behavior confirms the formation of lift due to the velocity difference between

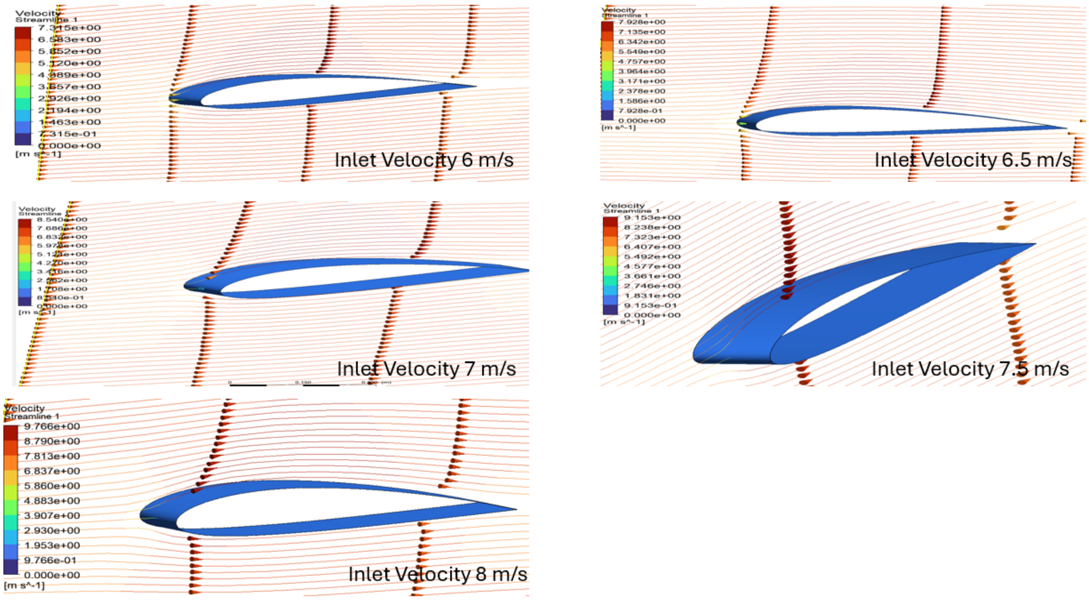

Figure 4.4: Static pressure contours of airfoil at different inlet velocities

Table 4.4: Comparison of Streamline Behavior at Different Velocities

Velocity (m/s)	Streamline Density	Flow Curvature	Wake Region	Flow Behavior
6	Low	Mild	Small	Smooth
6.5	Moderate	Slight	Slight	Smooth
7	High	Noticeable	Moderate	Attached
7.5	Very High	Strong	Larger	Stable
8	Maximum	Maximum	Largest	Stable

4.5 Summary

The CFD analysis of the airfoil was carried out at different inlet velocities ranging from 6 m/s to 8 m/s to evaluate its aerodynamic performance. The results from pressure contours, velocity magnitude, turbulent kinetic energy, and streamline patterns show a consistent and logical flow behavior around the airfoil.

The pressure distribution indicates that a high-pressure region is formed at the leading edge, while a low-pressure region develops over the upper surface, resulting in lift generation. As the inlet velocity increases, the pressure difference between the upper and lower surfaces increases, leading to enhanced lift.

The velocity magnitude results show that the flow accelerates over the upper surface of the airfoil, with higher velocities corresponding to lower pressure regions. The wake region behind the trailing edge becomes slightly more pronounced at higher velocities, but the flow remains attached.

The turbulent kinetic energy analysis reveals that turbulence is low near the leading edge and increases towards the trailing edge and wake region. Although turbulence intensity increases with velocity, it remains controlled and does not cause flow separation.

The streamline patterns confirm smooth and attached flow over the airfoil for all cases. Increased velocity results in higher streamline density and curvature, indicating stronger aerodynamic effects.

Overall, the results demonstrate that increasing inlet velocity improves the aerodynamic performance of the airfoil by increasing lift while maintaining stable flow with controlled drag characteristics.

CHAPTER 5

Conclusions and Future Scope

5.1 Conclusions

The present study focused on the CFD analysis of the aerodynamic performance of symmetrical and unsymmetrical airfoils under varying inlet velocities ranging from 6 m/s to 8 m/s. The investigation included detailed evaluation of pressure distribution, velocity magnitude, turbulent kinetic energy, and streamline patterns to understand the flow behavior and aerodynamic characteristics of both airfoil types.

From the pressure contour analysis, it was observed that both airfoils exhibit a high-pressure region at the leading edge and a low-pressure region over the upper surface, which is responsible for lift generation. However, the unsymmetrical airfoil produced a significantly higher pressure difference between the upper and lower surfaces due to its cambered geometry, resulting in enhanced lift even at lower velocities. The symmetrical airfoil, on the other hand, showed a more uniform pressure distribution and required higher velocities to achieve similar lift performance.

The velocity magnitude results indicated that the flow acceleration over the upper surface was more pronounced in the unsymmetrical airfoil. This increased acceleration corresponds to stronger suction and improved lift characteristics. In contrast, the symmetrical airfoil exhibited relatively balanced velocity distribution, contributing to stable flow behavior. The wake region behind the trailing edge became more noticeable at higher velocities for both airfoils, indicating increased mixing and energy dissipation.

The turbulent kinetic energy analysis revealed that turbulence intensity increases with inlet velocity for both airfoils, particularly near the trailing edge and in the wake region. The unsymmetrical airfoil showed slightly higher turbulence levels due to stronger velocity gradients and pressure differences.

Despite this increase, the turbulence remained controlled and did not lead to significant flow separation.

The streamline patterns confirmed smooth and attached flow over both airfoils across all velocity cases. The unsymmetrical airfoil displayed greater streamline curvature and density over the upper surface, indicating stronger aerodynamic effects and higher lift potential. The absence of recirculation zones or major flow separation in both cases indicates stable and efficient aerodynamic performance.

A combined analysis of all parameters shows that increasing inlet velocity enhances lift generation, increases turbulence intensity, and slightly enlarges the wake region. However, the flow remains attached, and aerodynamic efficiency improves as the increase in lift is more significant than the increase in drag.

In conclusion, the unsymmetrical airfoil demonstrates superior aerodynamic performance in terms of lift generation, making it suitable for applications where high lift is required, such as aircraft wings. The symmetrical airfoil, while generating lower lift at zero angle of attack, provides better stability and balanced flow characteristics, making it suitable for applications like control surfaces. The CFD results validate fundamental aerodynamic principles and highlight the effectiveness of numerical simulation in analyzing and predicting airfoil performance.

5.2 Future Scope

The present study focused on the CFD analysis of symmetrical and unsymmetrical airfoils at different inlet velocities under steady flow conditions. However, there are several areas where the work can be extended to obtain more comprehensive insights into aerodynamic performance.

Future work can include analysis at different angles of attack to study lift and stall characteristics more effectively. Investigating the performance under turbulent flow conditions using advanced turbulence models such as $k - \omega$ SST or Large Eddy Simulation (LES) can provide more accurate results. The study can also be extended to higher velocity ranges, including compressible flow regimes, to analyze transonic and supersonic effects.

Additionally, optimization of airfoil geometry using computational techniques can be carried out to improve lift-to-drag ratio. The effect of surface roughness, boundary layer control methods, and flow control techniques such as vortex generators can also be explored. Experimental validation of CFD results through wind tunnel testing can further enhance the reliability of the study.

5.3 Future Work

The present study can be further extended to gain deeper insights into the aerodynamic performance of symmetrical and unsymmetrical airfoils under varied flow conditions. Future investigations can focus on analyzing more complex flow phenomena and improving the accuracy of CFD predictions through advanced modeling techniques. In addition, exploring different operating conditions and design modifications can help in enhancing aerodynamic efficiency.

The following areas can be considered for future work:

- Analysis of airfoils at different angles of attack to study lift, drag, and stall characteristics
- Use of advanced turbulence models such as the $k - \omega$ SST model and Large Eddy Simulation (LES)
- Extension of the study to higher velocity ranges, including compressible flow regimes
- Evaluation of lift and drag coefficients (Cl and Cd)
- Optimization of airfoil geometry to improve lift-to-drag ratio
- Investigation of surface roughness and boundary layer control techniques
- Study of flow control methods such as vortex generators
- Validation of CFD results through wind tunnel testing

These extensions will help in improving the understanding of aerodynamic behavior and contribute to the development of more efficient and optimized airfoil designs for practical engineering applications.

5.4 Summary

The CFD analysis of the aerodynamic performance of symmetrical and unsymmetrical airfoils highlights the significant influence of airfoil geometry and inlet velocity on flow behavior. The results indicate that unsymmetrical airfoils produce higher lift due to stronger pressure differences and enhanced flow acceleration over the upper surface, whereas symmetrical airfoils exhibit more uniform pressure distribution and stable aerodynamic characteristics.

With increasing inlet velocity, both airfoils show improved lift generation, higher velocity gradients, and increased turbulence intensity. However, the flow remains attached in all cases, indicating stable aerodynamic performance without significant separation. Although drag increases due to higher turbulence and wake formation, the rate of lift increase is greater, resulting in improved aerodynamic efficiency.

Future scope of this work includes extending the analysis to different angles of attack to study stall behavior, implementing advanced turbulence models such as the $k - \omega$ SST model and Large Eddy Simulation (LES) for higher accuracy, and analyzing compressible flow conditions at higher velocities. Further improvements can be achieved through airfoil optimization, boundary layer control techniques, and experimental validation using wind tunnel testing. These advancements will enhance the understanding of aerodynamic behavior and contribute to the design of more efficient airfoil systems.

5.5 team work

Figure 5.1: Project team with guide during poster presentation of the CFD analysis of the aerodynamic performance of the NACA 2412 unsymmetrical airfoil

5.6 References

[1] I. H. Abbott and A. E. Von Doenhoff, *Theory of Wing Sections: Including a Summary of Airfoil Data.* New York, NY, USA: Dover Publications, 1959.

[2] J. D. Anderson, *Fundamentals of Aerodynamics*, 6th ed. New York, NY, USA: McGraw-Hill, 2017.

[3] J. Katz and A. Plotkin, *Low-Speed Aerodynamics*, 2nd ed. Cambridge, U.K.: Cambridge University Press, 2001.

[4] R. Ives, A. S. Keir, E. Bassey, and F. A. Hamad, "Investigation of the flow around an aircraft wing of section NACA 2412 utilising ANSYS Fluent," *INCAS Bulletin*, vol. 10, no. 1, pp. 95–104, 2018.

[5] M. S. G. Samuel and P. Rajendran, "CFD validation of NACA 2412 airfoil," Technical Report, 2019.

[6] M. S. G. Samuel, P. Rajendran, and S. A. Khan, "CFD simulation of NACA 2412 airfoil with new cavity shapes," *Advances in Aircraft and Spacecraft Science*, vol. 9, no. 2, pp. 131–148, 2022.

[7] Z. Gong, "CFD analysis of NACA2412 and NACA0012 at low Reynolds numbers based on ANSYS Fluent," *Applied and Computational Engineering*, vol. 127, no. 1, pp. 59–67, 2025.

[8] N. D. Baruah, D. Deka, and K. K. Das, "Aerodynamic analysis of NACA 2412 airfoil using ANSYS," *International Research Journal on Advanced Engineering and Management*, vol. 3, no. 2, pp. 173–177, 2025.

[9] M. R. Robby, A. Karudin, Refdinal, and Y. Fernanda, "Aerodynamic analysis of NACA 2412 airfoil on UAV aircraft wing using ANSYS Fluent software," *International Journal of Innovation in Mechanical Construction and Energy*, vol. 1, no. 3, 2024.

[10] S. A. Omenai, "CFD simulation of steady, compressible flow around the NACA 2412 airfoil," *International Journal of Computing and Engineering*, 2024.

[11] Y. Guo, "Aerodynamic performance and flow characteristics of the 2D NACA 2412 airfoil using CFD," *Applied and Computational Engineering*, 2026.

[12] O. J. E. Matsson, J. Voth, C. McCain, and C. McGraw, "Aerodynamic

performance of the NACA 2412 airfoil at low Reynolds number," in *ASEE Annual Conference and Exposition*, New Orleans, LA, USA, 2016.